



Yared Kinetibeb Tesfaye

5th-Year Computer Engineering Senior | Full-Stack, HFT & Embedded Systems Architect

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University: Addis Ababa Science and Technology University (AASTU)

EXECUTIVE SUMMARY

Highly accomplished 5th-Year Computer Engineering Senior student at Addis Ababa Science and Technology University (AASTU) specializing in high-frequency algorithmic trading systems, distributed Go microservices, AST code security auditors, and bare-metal IoT firmware. Proven track record of architecting sub-microsecond trading engines, 3D WebGL Digital Twins, and OWASP security scanners.

FEATURED ENTERPRISE ENGINEERING PROJECTS

FinPulse Engine — High-Frequency Algorithmic Trading & Risk Engine <i>Go 1.22 Python 3.11 Black-Scholes Options Monte Carlo VaR TradingView WebGL UI</i>	Architected sub-microsecond atomic lock-free ring buffer order queue and price-time priority L2 order book. Developed VWAP/TWAP institutional order slicer, Black-Scholes Options Greeks (Delta, Gamma, Vega, Theta), and 95% Monte Carlo VaR simulation engine.
NexusIoT Edge — Distributed Industrial IoT Telemetry Platform <i>Go 1.22 WebSockets Redis Pub/Sub Three.js 3D WebGL PostGIS Docker</i>	Engineered high-throughput telemetry ingestion microservice handling 100,000+ metric msgs/sec. Built 3D WebGL Three.js industrial turbine Digital Twin, multi-node comparative oscilloscope, and Web Audio siren warning annunciator.
SentinelAI — Autonomous Agentic AI Code Security & AST Auditor <i>Python 3.11 AST Compiler OWASP Top 10 ISO 27001 DevSecOps CI/CD</i>	Built multi-language AST parser detecting SQL Injection, API secrets, C++ buffer overflows, and Go panics. Integrated 6-axis security radar chart, live sandbox presets, and 1-click printable PDF ISO 27001 security certificate generator.
AASTU Academic Portal — University Management System <i>Go 1.22 RESTful API JWT Auth Middleware Weighted GPA Engine PostgreSQL</i>	Developed student profile API microservices featuring ECTS weighted cumulative GPA calculation algorithms, JWT auth middleware, and automated transcript PDF generation.

TECHNICAL SKILLS & COMPETENCIES

Languages & Core:	Go (Golang), Python 3.11, C / C++, TypeScript, JavaScript (ES6+), SQL, Embedded C, Dart
Backend & Cloud:	REST APIs, WebSockets, Redis Pub/Sub, Docker, PostgreSQL, Microservices, PlatformIO, FreeRTOS
Frontend & UI:	React 18, Three.js 3D WebGL, HTML5 Canvas API, TradingView Terminal Styling, Tailwind CSS

EDUCATION & CERTIFICATIONS

B.Sc. in Computer Engineering (5th-Year Senior) | AASTU (Cumulative GPA: 3.78/4.00)
Certifications: Simplilearn Generative AI Literacy • Udacity Full-Stack Web Developer • FreeCodeCamp Responsive Web Design